

the vulnerability issues. Everyone today, including the US government, is trying to spend more money in the detection of vulnerabilities and secure them. Working together as a community can allow us to solve these issues together.”

Dr Lokesh puts up a question to Gandhali Samant, “What is the main foundation and strength of open source which will play a key role in the metaverse?”

Gandhali responds, “Open source is both flexible and agile. Better talent keeps showing up and the maintenance cost is shared among people. It is important to note that the advantages of using open source far outweigh the disadvantages. GitHub is simply the largest open source community in the world, or you can say it is the home of open source itself. At GitHub our focus is to continuously improve the platform and add features so that the collaboration and open source projects can thrive.”

“Today, every new security feature that we build is available for free for public repositories. It is the main source of developing open source software. If you look up the top 10 metaverse projects on Google, you will find that they all are available on GitHub,” she adds.

“The role of GitHub in the metaverse will be very similar to the role that GitHub has played in the large-scale success and adoption of technologies like TensorFlow for AI or Docker or Kubernetes for containerisation, orchestration, Hyperledger, etc. GitHub will continue to provide the best platform for software development,” says Gandhali

Dr Lokesh further asks, “What would be the limitations and concerns for the collaborators?”

Gandhali states, “On one side we have multiple open source projects while on the other we have tech giants like Google and Microsoft that are already investing big dollars. No one company holds the monopoly for any

component today. They are bound to come together and work as each of them brings a different value to the table. We would see more standardisation across some of the underlying fundamental blocks to ensure compatibility. For instance, Linux started as an open source project while Microsoft Windows was totally closed source, yet they still have too many partnerships.”

Dr Lokesh puts up a question to Dr Sumit, “What are your views on data privacy with respect to the information protection aspects?”

Dr Sumit responds, “We are already hearing a lot of copyright and patent infringement cases around this as many open source projects are being used without the consent of the original creator. Although we have a better control over such scenarios in the real world, it is not the case in the metaverse as the person is not physically present there. We are not yet sure if the virtual property in the metaverse will have any analogy in the physical world.”

Dr Lokesh asks Sayed Mohammad Khasim, “Can you elaborate on the hardware aspect of it, the challenges that you face, and the elements that are already in place?”

“Technology today has evolved exponentially in terms of security, power, and features. Every smart device today is connected to a network, which is also exposing them to various security threats. This has made semiconductor companies put a lot of their focus on resolving these security threats,” responds Sayed Mohammed Khasim.

He adds, “We have an initiative in Texas Instruments where our software is completely giving you the accelerated AI with the DSPs, which is opened to the public.”

Dr Lokesh goes on to ask Biju Nair, “How do you see the landscape changing from the physical to virtual world since nothing is free?”

Biju Nair responds with, “Let

us say you have a trademark in the physical world but you have no presence in the virtual world, then the chances of you getting an injunction trademark misuse is very low. Such a case has already been brought before a US court by the name E.S.S. Entertainment versus Rock Star Videos. It stated that the depiction of the trademark of any video game does not influence the trademark of the owner as the video game is an artistic expression protected by the first amendment. The consumers would not be confused between the trademark produced by these sophisticated video games. So you may not be protected in the metaverse, unless you have some sort of presence over there.”

Dr Lokesh asks another question to Biju, “What are some of the best practices for the businesses from the legal perspective while they are collaborating in the context of open source?”

Biju Nair states in his response, “First, companies need to revisit their existing agreements and see what rights have been provided to them in those agreements and plan accordingly. Second, we will check if they have filed for the trademark for virtual goods or not, because if you won’t do that then you will be at a risk of losing the rights for those goods in the metaverse workspace.”

Dr Lokesh mentions at the end, “There are a lot of legalities that we need to look at and address them separately. Nonetheless, it is going to be pretty interesting. There is still a lot to be explored and developed in the world of metaverse.” **EFY**

This article has been put together from a panel discussion at Open Source Conference 2022: Proliferation of Open Source for Emerging Technologies, organised by Samsung R&D Institute India – Bangalore and IEEE ComSoc Bangalore Chapter. The panelists were Dr Lokesh Boregowda (Samsung), Gandhali Samant (GitHub India), Syed Mohammed Khasim (Texas Instruments), Biju K. Nair (LegalTech), Dr Sumit Kalra (IIT Jodhpur), and Vinay G. Rajagopal (Red Hat). It has been transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY with a knack for open source exploration and research.